
Clustering Indonesian companies' Annual Reports: preliminary assessment of the implementation of integrated reporting by Indonesian listed companies

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Abstract: The purpose of this research is to explore and analyse whether Indonesian companies' Annual Reports already disclose matters according to the Integrated Reporting (IR) Framework. This research analyses eight IR components. The research method employed is exploratory descriptive. The population is 493 companies listed on the Indonesian Stock Exchange that issued an Annual Report for the year 2014. The data were collected by browsing reports on the companies' websites and checking for disclosure using a checklist consisting of 55 items. The research uses cluster analysis and shows that, on average, Indonesian companies disclosed 53.81% of the IR components in their Annual Reports. This research also classifies the companies into four clusters, with Cluster 3 having the highest disclosure level, and Cluster 4 the lowest. Clusters 2 and 1 have moderate results, and the majority of companies belong to these clusters.

Keywords: Annual Report; business sectors; disclosure; integrated reporting.

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1 Introduction and literature review

Reporting is a way to communicate information to stakeholders. Currently, many forms of reporting exist. Besides traditional financial disclosure, environmental and CSR disclosures began to be made towards the end of the 20th century. Murdayanti, Gurendrawati and Iyabu (2014) stated that the purpose of disclosure is to submit a report to serve a variety of parties who have different interests. Although many forms of reporting exist, there is still doubt about the value of the information contained in a report. Purnomo and Widianingsih (2012) indicated that CSR disclosure is unable to strengthen the association between environmental performance and financial performance. The Federation of European Accountants (2015) stated in its report that corporate reporting is evolving and is now facing several issues, one of which is the absence of a single comprehensive report that summarises a company's affairs. To meet the great demands of stakeholders, the International Integrated Reporting Committee (IIRC) issued the integrated reporting (IR) Framework in 2013.

IIRC (2013) defined integrated reporting as a concise communication about how an organisation's strategy, governance, performance and prospects, in the context of its external environment, will lead to the creation of value over the short-, medium- and long-term. IIRC (2013) also stated that a full integrated report must have nine components, covering:

- 1 organisational overview and external environment
- 2 governance
- 3 business model
- 4 risks and opportunities
- 5 strategy and resource allocation
- 6 performance
- 7 outlook
- 8 basis of preparation and presentation, and (in doing this, taking into account)
- 9 general reporting guidance.

There are several guiding principles that must be followed when preparing an integrated report, namely:

- 1 strategic focus and future orientation
- 2 connectivity of information
- 3 stakeholder relationships
- 4 materiality

- 5 conciseness
- 6 reliability and completeness
- 7 consistency and comparability.

Currently, several countries already require companies to adopt Integrated Reporting or at least urge them to do so. South Africa has already required all listed companies to use Integrated Reporting, based on the King Code III. Several companies in Australia elected to adopt Integrated Reporting early, while France and UK already require companies to issue integrated sustainability and financial reports, for which the framework is similar to the IIRC integrated framework. Germany already requires combined financial and sustainability reporting but has yet to mandate Integrated Reporting.

Current research has proved that IR will be useful for financial decision-making. Bernardi and Stark's (2015) research concluded that IR results in environmental, social and governance (ESG) disclosure being associated with forecast accuracy. Research conducted by Zhou, Simnett and Green (2015) on companies listed on the Johannesburg Stock Exchange also showed that IR is providing the capital markets with information that is incrementally useful over existing reporting mechanisms. Some research has also suggested that IR implementation is not easy. Bavagnoli et al. (2014) suggested that companies still interpret 'materiality' and 'conciseness' in IR in different ways. Stubbs and Higgins (2014) stated that the adoption of IR does not trigger a radical or innovative way of disclosing items, but rather is just a form of reporting that is incrementally different from the previous sustainability reporting. Stubbs et al. (2014) conclude that implementing IR is a tough challenge, and that the lack of specific guidance is one factor that means that financial capital providers will not understand much of the detail about IR. Van Zyl (2013) also found that although many companies are attempting or claiming to be creating Integrated Reports, the level of integration and the understanding of what a truly Integrated Report should represent is still very low. Abeysekera (2013) stated that IR brings new challenges to auditors, and that auditors need detailed information about the assumptions for IR to help them develop an informed opinion about information outside financial reports. Levels of IR disclosure can vary depending on business sectors. Serafeim (2015) indicated that companies whose investors are more active in environmental and social issues may practise IR more consistently.

Several countries implement IR by simply requiring companies to combine financial information and non-financial information. Research conducted by Hao (2014) has indicated that it is necessary to view Integrated Reporting as a staged process, which evolves from combined reporting, to financial integration reporting and eventually to holistic reporting. As of 2016, Indonesia had not obliged or persuaded its companies to adopt or implement full IR, but Indonesian companies already publish Annual Reports that combine financial and non-financial information. It is very interesting to analyse whether Indonesian companies' Annual Reports already disclose what is required by the IR components, to give a preliminary assessment of what 'gaps' will have to be filled by Indonesian companies once they are required to apply IR. The purpose of this research is to explore the disclosure of the IR components in Indonesian companies' Annual Reports. This research will be useful for companies, so that they can assess whether there are any gaps between their Annual Reports and the IR Framework, and for regulators, to allow them to assess how near Indonesian companies are to implementing IR. It can also help with formulating work programs to help companies to implement IR.

2 Research methodology

This research employed an exploratory descriptive method. Since research about the process of adopting IR by examining Annual Reports in Indonesia is new, and research about IR is still limited, this is the best approach to adopt. The research population is all companies listed on the Indonesian Stock Exchange (the IDX) in 2014, a total of 521 companies. This research will describe eight out of the nine IR components. We decided to exclude the ninth component, general reporting guidance, since the assessment of this component requires heavy judgements to be made by a researcher, and these could be biased and subjective.

We collected the data by downloading Annual Reports from companies' and the stock exchange's websites and conducting content analysis by making a disclosure checklist of 55 items that have to be disclosed under the IR Framework. The checklist indicated whether the IR component was disclosed or not. The company scored 1 if the item was disclosed, or 0 if it was not. The total disclosure score was calculated by adding together the individual disclosure checklist items. The disclosure checklist used in this research can be described as follows (Table 1):

Table 1 Integrated reporting components checklist

<i>Items to be disclosed</i>	<i>Number of items on checklist</i>
Organisational overview and external environment (Org_Ovw)	11
Governance (Gov)	9
Business model (Bus_Mod)	10
Risks and opportunities (RO)	4
Strategy and resource allocation (Strat_RA)	8
Performance (Perf)	4
Outlook (OL)	3
Basis of preparation and presentation (BPP)	4
Total	55

We analysed the data in several steps. First, we described the data for IR component disclosure according to the eight IR components and the nine sectors of companies in IDX, namely:

- 1 Agriculture (AGRI)
- 2 Mining (MINING)
- 3 Basic Industry and Chemicals (BASIC-IND)
- 4 Miscellaneous Industry (MISC-IND)
- 5 Consumer Goods Industry (CONS)
- 6 Property, Real Estate and Building Construction (PROP)
- 7 Infrastructure, Utilities and Transportation (INFST)
- 8 Finance (FIN)
- 9 Trade, Services and Investment (TRADE). Second, we clustered the companies into several clusters using cluster analysis.

Cluster analysis is a multivariate analysis used to classify members of populations into groups with similar characteristics (Hair et al., 2005). We assigned the companies into clusters using the IR components as the characteristics of the clusters. To determine the number of clusters to be formed, hierarchical analysis using Ward's method was used, and the analysis proceeded using non-hierarchical analysis and the *K*-means method to distribute the companies into clusters. We also conducted a multicollinearity test and an *F*-test to check the validity of the cluster model.

3 Results and analysis

3.1 Data collection and descriptive statistics

From 521 companies, we found 493 Annual Reports (94.63%) from either the IDX website or the company's website. For the remaining 18 companies (5.37%), the Annual Report was unavailable on the IDX website or the company's website, and/or there was a broken link or website issues. The results for the overall IR component disclosure from the Indonesian companies' Annual Reports classified by IDX business sector can be presented in Table 2 as follows:

Table 2 Descriptions of IR component disclosure

<i>Industrial sector</i>	<i>IR component disclosures</i>								<i>Sector average (%)</i>
	<i>Org_Ovw (%)</i>	<i>Gov (%)</i>	<i>Bus_Mod (%)</i>	<i>RO (%)</i>	<i>Strat_RA (%)</i>	<i>Perf (%)</i>	<i>OL (%)</i>	<i>BPP (%)</i>	
AGRI	78.18	49.44	45.00	93.75	40.63	50.00	83.33	23.33	55.36
MINING	79.67	46.20	24.68	79.61	35.54	35.53	86.84	36.84	51.72
BASIC-IND	75.51	48.75	30.00	80.65	21.57	4.84	59.14	32.80	43.72
MISC-IND	69.00	52.71	30.77	69.87	33.65	8.97	67.52	33.33	45.97
CONS	71.25	68.47	40.27	64.19	50.34	35.81	80.18	49.10	57.10
PROP	81.48	62.68	53.40	86.32	37.50	58.49	49.06	44.03	59.73
INFST	80.87	79.63	63.13	91.15	39.84	46.35	65.97	60.42	66.67
FIN	74.19	49.17	49.31	89.08	46.98	26.44	61.69	50.96	56.01
TRADE	72.73	61.57	38.07	81.88	37.39	22.48	46.79	40.52	51.54
Average	75.32	57.93	41.90	82.11	37.94	28.87	61.54	43.06	53.81

Note: For presentation in the table, the scores have been converted into percentages.

The result in Table 2 shows us several interesting facts. First, overall the IR component disclosure in Indonesian companies' Annual Reports is 53.81%, or on average, 26 items out of 49 were disclosed. This also means that, on average, Indonesian companies only disclosed half of the information stipulated by the IR framework. Second, of the eight IR components analysed, Indonesian companies made most disclosures under the risk component, with an average of 82.11%, and they made fewest disclosures under the performance component, with an average of 28.87%. Indonesian companies are normally required to disclose the key risks and opportunities for their business and to analyse the impact of these on their business, so the disclosure rate for risk is high. The performance

component scored low, because most of the companies only disclosed financial matters, as required by the accounting/regulatory standard. Third, of the nine Indonesian Stock Exchange sectors analysed, companies in the Infrastructure, Utilities and Transportation sector disclosed most items, with an average of 66.67%, while companies in the Basic Industries and Chemicals sector disclosed the fewest items, at an average of 43.72%. The results show that Basic Industries companies only scored 4.84% on the performance component, the lowest score over all sectors.

3.2 Cluster analysis

Based on the multicollinearity test conducted using a statistical software package, all the variance inflation factors (VIF) were between 0.1 and 10, indicating that multicollinearity did not exist in the data. ANOVA analysis showed that all the IR components varied significantly in each cluster, with a *p*-value less than 0.05. The ANOVA tables and calculations are as follows (Table 3):

Table 3 Multicollinearity and ANOVA test results

<i>IR component</i>	<i>VIF</i>	<i>F</i>	<i>Significance value (p-value)</i>
Org_Ovw	1.651	151.175	0.000
Gov	1.596	128.547	0.000
Bus_Mod	2.132	163.970	0.000
RO	1.462	189.219	0.000
Strat_RA	1.442	64.458	0.000
Perf	1.498	81.937	0.000
OL	1.192	126.577	0.000
BPP	1.461	144.341	0.000

The next step is to determine the clusters for classifying the Annual Reports of the 493 companies according to the eight IR components. Before conducting the analysis using the Ward's method, the data were transformed into *z*-scores, to eliminate problems associated with the variability of data. The analysis using the Ward's method is shown in Table 4:

Table 4 Hierarchical analysis cluster forming using Ward's method

<i>Number of clusters</i>	<i>Agglomeration coefficient</i>	<i>Decrease in coefficient (%)</i>	<i>Decreasing rate (%)</i>
1	3.936.000	–	–
2	2.978.241	24.33	24.33
3	2.490.848	1637	7.97
4	2.295.059	7.86	8.50
5	2.130.830	7.16	0.70
6	1.995.009	6.37	0.78

Note: The numbers of clusters to be formed using Ward Method will be determined from decreasing rate of coefficient (%) and decreasing rate (%). Since the decreasing rate of coefficient and decreasing rate from cluster 5 onward is relatively small, so the cluster formed should be four.

Based on the Ward's method, we can see that the decreasing rate of the agglomeration coefficient from one cluster to four clusters is significant, but from five clusters onwards, the decreasing rate is relatively stable. Based on the Ward's method, the number of clusters to be formed is determined as four.

3.3 *Final cluster solutions and discussion*

After determining the number of clusters, the next step is to divide the 493 companies into four clusters. After processing the data using the K-means method, the final clusters, with descriptions of the mean of the rate of disclosure of the items in each clusters (please see Table 5 for detail descriptions) are as follows:

Table 5 Final cluster descriptions

<i>Descriptions</i>	<i>Cluster name</i>			
	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
Org_Ovw	8.313	8.867	9.233	6.935
Gov	4.755	4.842	7.243	4.293
Bus_Mod	4.000	4.683	6.602	1.943
RO	3.741	3.667	3.757	2.016
Strat_RA	2.864	3.658	4.029	1.797
Perf	0.912	1.358	2.175	0.447
OL	1.272	2.667	2.233	1.439
BPP	2.259	2.350	3.728	2.179
Ranking of cluster (1st is the best, while 4th is the worst)	3rd	2nd	1st	4th
Total number of companies	147	120	103	123

The profile of each cluster can be described as follows:

- 1 Cluster 1 consists of 147 companies. This is the cluster that includes the highest number of companies. This cluster has an overall IR component score lower than that for cluster number 2, but there is more extensive disclosure on risk. Disclosure on outlook is the lowest among the clusters. Companies in this cluster are predominantly in the Trading sector, followed by the Financial sector and the Property sector.
- 2 Cluster 2 consists of 120 companies. This cluster has an overall IR component score lower than that of cluster number 3, but the disclosure on outlook is the best among all the clusters. Disclosure on risk is the second lowest among all the clusters. Members of this cluster are mostly in the Mining sector, followed by the Financial and Agricultural sectors.
- 3 Cluster 3 consists of 103 companies. This is the cluster with the lowest total number of companies. This cluster has the highest overall IR component score among all the clusters, with an exception only for the outlook component, which is the second highest. Members of this cluster belong mostly to the Infrastructure sectors, followed by the Financial and Property sectors.

- 4 Cluster 4 consists of 123 companies. This cluster has the lowest overall IR component score among all the clusters, with an exception only for the outlook component, which is the second lowest. Members of this cluster predominantly come from the Basic Industries sector, followed by the Trade and Miscellaneous Industry sectors. Table 6 showed number of companies according to sectors in each clusters.

Table 6 Sectors of companies in each cluster

Cluster number	Total number of companies per sector								
	AGRI	MINING	BASIC-IND	MISC-IND	CONS	PROP	INFST	FIN	TRADE
1	3	1	22	11	5	22	4	23	56
2	15	24	12	9	10	10	6	24	10
3	0	2	2	1	10	18	33	23	14
4	2	11	26	18	12	3	5	17	29

4 Conclusion and suggestions

This research gives a preliminary assessment of Indonesian companies' progress towards adopting Integrated Reporting. The research shows that Indonesian companies only disclosed 53.81% of the components described in the IR framework. The difficulty of adopting Integrated Reporting will vary across business sectors. Companies in some sectors, like the Infrastructure sector, disclosed more than companies in other sectors, while companies in sectors like Basic Industries and Miscellaneous Industries disclosed less than 50% of the components. Companies in these sectors need to make a great improvement in their reporting disclosure, while other industries also need to improve, since the disclosure levels range from 50% to 67%. Cluster analysis has already showed us that out of the four clusters, Cluster 3, which has the best disclosure level, only contains 103 companies from the total of 493 companies. Cluster 4, which has the worst disclosure level, contains 123 companies, which is higher than the number in Cluster 3. The majority of companies were classified in Cluster 1, which has the second lowest disclosure rate. This result shows that companies in Indonesia still need to improve their reporting disclosure.

Finally, this research is exploratory analysis, so it still needs improvement in the future. Using cluster analysis has limitations in its results, descriptions and interpretations. If Indonesia was to adopt Integrated Reporting, then further research could find more factors that influence or determine the level of adoption of Integrated Reporting. Further research can also consider using a combination of quantitative and qualitative techniques, such as interviews in companies to assess the difficulties in adopting Integrated Reporting, or interviews of regulators regarding how the regulators are reacting to the process of the adoption of Integrated Reporting.

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